

Introduction

GARCH (Generalised Autoregressive Conditional Heteroscedasticity) models capture time-varying conditional variance — the volatility clustering pattern in which quiet periods alternate with volatile ones. Financial returns are the prototypical application: stock prices show stretches of low daily variability punctuated by high-volatility crisis episodes, a structure that conditional-Normal ARMA models cannot accommodate. GARCH extends ARMA by adding an autoregressive equation for the conditional variance itself, making volatility a forecastable quantity.

Prerequisites

A working understanding of ARMA models, conditional distributions, and the distinction between unconditional and conditional variance.

Theory

The GARCH(1, 1) model decomposes returns as

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \sim \mathcal{N}(0, 1),$$

with the conditional variance evolving as

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

The mean structure can be combined with an ARMA component; GARCH specifically models the variance of the innovations. Stationary unconditional variance requires $\alpha + \beta < 1$; values close to 1 indicate highly persistent volatility (typical for daily financial returns).

Extensions handle asymmetric responses to positive vs negative shocks (EGARCH, TGARCH, APARCH), variance feedback into the mean (GARCH-M), and multivariate volatility (DCC, BEKK). For non-Normal returns, t - or skewed- t -distributed innovations are standard.

Assumptions

Conditional Normality (or t , skewed- t) of innovations z_t ; stationarity of the conditional variance ($\alpha + \beta < 1$); the chosen GARCH order is appropriate for the data.

R Implementation

```
library(rugarch)

# Simulate GARCH(1,1) returns
spec <- ugarchspec(variance.model = list(model = "sGARCH", garchOrder = c(1, 1)),
  mean.model = list(armaOrder = c(0, 0), include.mean = TRUE),
  distribution.model = "norm")

set.seed(2026)
sim <- ugarchpath(spec = ugarchspec(
  variance.model = list(garchOrder = c(1, 1)),
  mean.model = list(armaOrder = c(0, 0))),
  n.sim = 500)
r <- as.numeric(fitted(sim))

# Fit
```

```
fit <- ugarchfit(spec, data = r)
show(fit)
plot(fit, which = 3)    # conditional sd
```

Output & Results

`ugarchfit()` returns parameter estimates (ω, α, β , mean parameters), log-likelihood, AIC/BIC, and Ljung-Box-style tests on standardised and squared standardised residuals. The conditional-standard-deviation plot shows the time-varying volatility extracted by the model.

Interpretation

A reporting sentence: “A GARCH(1, 1) fit to daily returns estimated $\hat{\omega} = 0.0001$, $\hat{\alpha} = 0.08$, $\hat{\beta} = 0.91$ ($\hat{\alpha} + \hat{\beta} = 0.99$, very persistent volatility); standardised residuals showed no remaining ARCH effects (Ljung-Box on squared residuals $p = 0.42$). Conditional-volatility forecasts widen modestly over the 30-day horizon.” Always report the persistence sum and standardised-residual diagnostics.

Practical Tips

- Check standardised residuals (`residuals(fit, standardize = TRUE)`); they should be approximately iid and often heavier-tailed than Normal, motivating t or skewed- t innovations.
- For asymmetric responses to positive vs negative shocks (the leverage effect in equity returns), use EGARCH or TGARCH; standard GARCH is symmetric and misses this.
- `rugarch` is the industry standard for univariate GARCH; for multivariate volatility (covariance matrices), `rmgarch` extends to DCC, BEKK, and copula-GARCH.
- Value-at-Risk forecasts come from the conditional-variance forecast plus a quantile of the innovation distribution; `ugarchforecast()` and `ugarchroll()` support rolling VaR estimation.
- GARCH on returns is conventional, but the framework applies to any series with volatility clustering — physiological signals, atmospheric data, traffic counts.
- For very long series, the parameter estimates can be stable but the conditional-volatility series may need recursive re-estimation; use `ugarchroll()` for genuine out-of-sample evaluation.

R Packages Used

`rugarch` for univariate GARCH and its many extensions; `rmgarch` for multivariate GARCH (DCC, BEKK); `fGarch` as an alternative implementation; `bayesGARCH` for Bayesian GARCH; `tsgarch` for the modern tidyverse-flavoured interface.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Time series basics